



Modelling and optimization of methanol synthesis from hydrogen and CO₂

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ABSTRACT

In this paper, CO₂ hydrogenation to methanol is investigated with three objective functions, i.e. maximization of reactor productivity, CO₂ utilization, and hydrogen conversion. Key performance indicators of methanol production via conventional and CO₂ hydrogenation routes are compared.

The results suggest that the methanol production rate with an inlet H₂/CO₂ of 3 is ca. 59 % greater than the case with the inlet H₂/CO₂ composition of 2. The overall hydrogen and CO₂ conversion of the cases with the inlet H₂/CO₂ of 3 is greater than the reactor system with the inlet H₂/CO₂ of 2.

For each objective function criteria, the analysis shows that increasing the H₂/CO₂ to the methanol reactor increases the natural gas consumption in the conventional route and the higher the inlet H₂/CO₂ ratio, the higher the CO₂ emission is. If the inlet hydrogen is produced via water electrolysis, the CO₂ emission of the CO₂ hydrogenation process route will be negative.

1. Introduction

Methanol with chemical formulae of CH₃OH is historically known “wood alcohol” and is the simplest alcohol (methyl alcohol) which is colorless, volatile, light, and flammable liquid. Energy density of methanol is 4.4 kW h/l (5.5 kW h/kg) and the volumetric density of methanol is almost 6 times higher than that of hydrogen.

Noble Prize winner, George A. Olah, introduced the concept of methanol economy in 1994. The global methanol demand is currently about 70 million tonne per year with a growth rate of 6–8% per year and it is anticipated to reach over 90 million tonne per year in the coming years [1]. Methanol is an essential ingredient with a wide range of applications. More than 50 % of methanol demand is used as feedstock for production of propylene/poly-propylene, acrylic acid/acrylates, ethylene/propylene, hydrogen, acetic acid, formaldehyde, etc. Methanol can also be used in gasoline blending. Other applications include use of methanol in fuel cells and for power production. The global oil and gas dependence could be lessened by substituting these fuels with methanol. Fig. 1 presents the breakdown of methanol consumption by final application. The highest methanol demand is for the production formaldehyde (30 %) followed by MTBE (12 %), fuels (12 %), acetic acid (10 %). About 6% of the methanol is used to produce DME.

The greenhouse gases (GHGs) accumulation in the atmosphere and its impact on global warming is a critical issue and we must deal with it. The most important gasses ranked by the direct contribution to the greenhouse effect are water vapor (36–70 %), CO₂ (9–26 %), CH₄

(4–9%) and ozone O₃ (3–7%), [2]. The Global Warming Potential (GWP) of each kilogram of CH₄ is approximately 84 times higher than one kilogram of CO₂ for a 20-year period. Currently, the total CO₂ emissions are about 32.2 billion tonne per annum [3]. The electricity and heat production sector with a share of about 42.4 % has the highest contribution to the global CO₂ release.

Methanol is a long-term chemical energy storage option to store the intermittent energy from renewable sources. The excess energy and/or intermittent renewables can be used to produce hydrogen by electrolyzing water. The produced hydrogen then reacts with CO₂ to produce methanol. There are numerous sources of CO₂ to be used for CO₂ hydrogenation to methanol e.g. flue gas incineration, biogas purification, ethanol fermentation, steel mills, lime and cement industry, air capture, coal- and gas-fired power plants, etc. Owing to the increasing awareness of the obligations to mitigate CO₂ emissions and because of government politics (e.g., cap-and-trade programs and CO₂ tax), the conversion of CO₂ to methanol gains industrial relevance.

The global CO₂ conversion into value-added chemicals such as (urea, methanol, di-methyl-ether (DME), methyl tert-butyl ether (MTBE), formaldehyde, biodiesel, etc.) is about 0.2 billion tonne per annum [2]. The global CO₂ demand for methanol and DME synthesis is respectively 8 million tonne per year and nearly 3 million tonne per year [2]. A comprehensive review on physical and chemical CO₂ valorization pathways can be found in Ref. [2]. Utilization of CO₂ in methanol production can either be indirectly (in the reforming stage) or directly in the methanol reactor. For example, Fanavaran methanol plant with annual capacity of one million tonne receives natural gas

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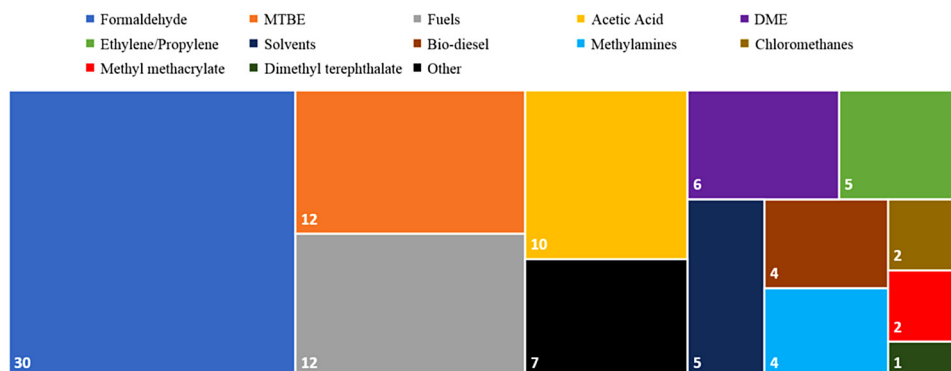


Fig. 1. Global methanol demand by derivative (Percent).

(610 kilo tonne) and CO₂ (268 kilo tonne) from two petrochemical complexes. The natural gas, CO₂, and steam are fed to a steam methane reformer (SMR) and the produced synthesis gas (syngas) is then conveyed to a downstream methanol synthesis reactor.

The literature is rich in CO₂ utilization for methanol synthesis. For example, four different system configurations including 1) integrated coal-to-liquids (CTL) and natural gas-to-liquids (GTL) plants, 2) combined nuclear-derived energy with CO₂ hydrogenation to methanol process or CTL, 3) use of dry methane reforming (DMR) for chemicals and fuels production, and 4) integrated solar energy with H₂O–CO₂ splitting for chemicals and fuels production was analyzed in Ref. [4]. The combination of the processes using fossil fuels feed with carbon-free energy source reduces the amount of the CO₂ release to the environment. Machado et al. investigated CO₂ hydrogenation and methanol production route from syngas [5]. The effect of process conditions was studied. The CO₂ consumption was 665 kilo tonne per year and the CO₂ emission was 19.02 t per hour in the CO₂ hydrogenation route. The feasibility of offshore CO₂ utilization by DMR in methanol synthesis was studied in Ref. [6] from a techno-economic and environmental viewpoint. CO₂ valorization through methanol synthesis was studied in Refs. [7,8]. Six syngas production routes including DMR with CO removal, DMR with H₂ addition, SMR, bi-reforming, tri-reforming, and a parallel arrangement of DMR and SMR were considered. The DMR route with H₂ addition results in a lower CO₂ emission per unit of mass of the methanol produced. In another work, Pisarenko and colleagues [9] developed a methanol production process where syngas was produced by DMR. The process encompasses four serial methanol reactors. The results suggest that CO₂ recycling to the reforming section results in an increased methanol production. The power to methanol concept developed by Carbon Recycling International (CRI) company refers to the case that power is stored in methanol. The largest CO₂ hydrogenation plant is located in Iceland [10]. This methanol plant uses 5.5 kilo tonne CO₂ per annum. The energy requirements of the methanol plant are supplied by hydropower and geothermal energy. Hydrogen is provided by water electrolyzers.

Chakraborty et al. reviewed various routes for the anthropogenic CO₂ capture and further conversion to methanol [11]. The membrane-integrated photocatalytic approach is a green route to have high methanol yield and very high methanol purity. Samimi and co-workers investigated the methanol synthesis from hydrogen and CO₂ [12]. Syngas was first produced in a reverse water gas shift reactor and then the syngas was conveyed to a methanol reactor. The methanol reactor was equipped by water perm-selective membrane to remove water from the reactor. The methanol production rate of the process is nearly 21 % higher than the conventional methanol synthesis route. In another study, two pathways for indirect CO₂ utilization for methanol production via reverse water gas shift reaction were studied in Ref. [13]. In the first route, CO₂ was first converted to syngas in a reverse water gas shift reactor, was fed to a condenser, and then was conveyed to the methanol production reactor. In the second pathway, a water permselective

membrane was used in the reverse water gas shift reactor. The results suggested the superiority of pathway 2 over pathway 1, owing to the following reasons: a higher CO yield and CO₂ conversion, a more appropriate syngas composition, about 4.15 % higher methanol production rate, and ca. 17 % lower water production. Samimi et al. studied the phase stability of CO₂ hydrogenation to methanol in gas cooled, water cooled, and double (water + gas) cooled tubular reactors [14]. The results suggest that the gas cooled reactor layout has higher methanol production and CO₂ conversion. In addition, liquid phase does not form in the gas cooled layout. Various CO₂-based methanol production routes were analyzed by Gonzalez-Garay et al. [15]. The results show that production of green methanol is economically unattractive due to its higher cost compared to the conventional fossil-based route. In Ref. [16], Li et al. reviewed the recent activities for producing renewable H₂ and CO₂, in addition to the developments in the production of bimetallic catalysts for CO₂ hydrogenation to methanol.

In this study, possible design advances for the CO₂ hydrogenation to methanol (as an energy storage option) through a fixed-bed reactor is investigated to maximize methanol production rate, CO₂ conversion (CO₂ utilization), and hydrogen conversion.

2. Methanol production process

2.1. CO₂ Hydrogenation to methanol process

The conversion of CO₂ and hydrogen to methanol process removes the costly syngas production section of the conventional methanol production pathway (from fossil fuels) if the hydrogen feed is produced from renewables [17].

Various types of reactor including adiabatic, continuous stirred-tank (CSTR), fluidized bed, gas-, or water-cooled fixed bed reactors can be used for methanol synthesis [18,19]. Adiabatic and gas-cooled reactors demonstrate fairly low and medium fixed capital costs. CSTR reactors have lower conversion and operating temperature. The methanol production rate of fluidized bed reactors are higher than the fixed bed reactors [20]. Methanol synthesis in fixed-bed tubular reactors is a common industrial practice. The pressure of the coolant medium flowing in the shell side facilitates the control of the methanol reactor temperature by the water phase change.

The CO₂ hydrogenation process to methanol includes water-cooled multi-tubular fixed bed reactors (Fig. 2a). The CO₂ can be imported over the fence from CO₂ emitting industries. The hydrogen stream, on the other hand, can be produced by electrolyzing excess power to H₂ and O₂. Methanol synthesis is performed at pressure range of 50–80 bar and moderate temperature range of 200–300 °C enclosed by a shell. The recycling of unconverted syngas can be done in the methanol synthesis owing to the fact that reactions conversion is limited by thermodynamic equilibrium.

The decision variables of the reactor system of Fig. 2a are the coolant medium temperature, heat transfer area, H₂/CO₂ molar ratio to

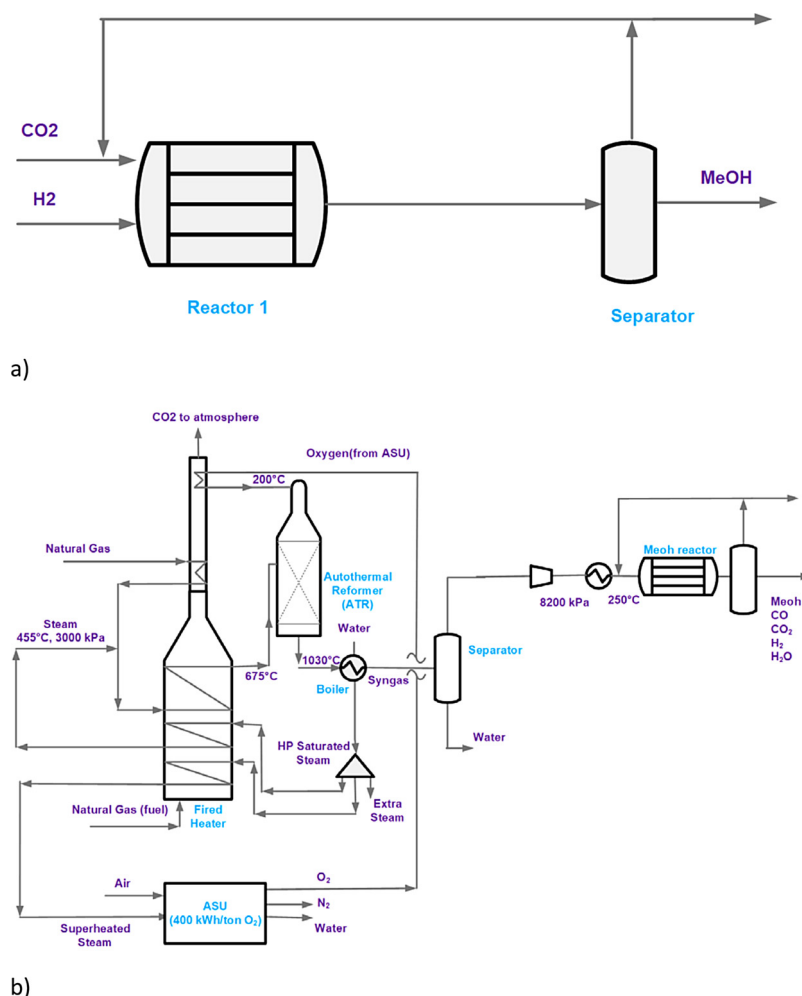


Fig. 2. a) Methanol synthesis process from CO₂ and H₂, b) Conventional methanol production route from natural gas where natural gas is first converted to syngas and then to methanol.

the reactor, and level of catalyst dilution. For a given feed composition and flowrate, different objective functions can be considered, e.g. maximization of CO₂ utilization, methanol production rate at the effluent of the reactor, hydrogen conversion, etc.

2.2. Conventional methanol production process

Conventional methanol production routes use a two-step catalytic chemical process i.e. syngas production from natural gas, syngas conversion to methanol, and then methanol purification. Various reforming technologies such as steam methane reforming (SMR), dry methane reforming (DMR), auto-thermal reforming (ATR), and the like can be used to convert fossil fuels e.g. natural gas to syngas [21]. Other feedstocks such as biomass, coal, and municipal waste could also be used to produce syngas. Currently, the estimated capital investment costs of most large-scale methanol production plants are \$500–\$700/tonne MeOH/year with an average value of about \$530 /tonne MeOH/year [22]. In addition, the estimated capital investment costs of small-scale methanol production plants may vary from \$700 to \$1100/tonne MeOH/year.

Fig. 2b represents conventional methanol production from natural gas. As evident from this figure, natural gas is preheated in the fired heater and then is mixed with steam and enters the auto-thermal reformer (ATR) at 675 °C [23–25]. Pure oxygen from the air separation unit (ASU) is heated up to 200 °C and is fed to the ATR [26–28]. In the ATR, natural gas is converted to syngas which mainly consists of

hydrogen and CO [29,30]. The hot syngas leaving the ATR is cooled down to separate water from it, produce steam and then is compressed before entering the methanol reactor.

3. Mathematical modelling of methanol reactor

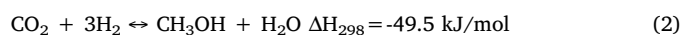
The main assumptions used to develop the reaction path model are one-dimensional flow, steady-state mode of operation, gas phase reactions, no extra feed distribution, water-cooled reactor, plug flow model for CO₂ hydrogenation to methanol, the feed temperature is 220 °C, reactor pressure is fixed at 82 bar, reactor temperature is constrained by 270 °C, and the total volume (*V*) is 100 m³.

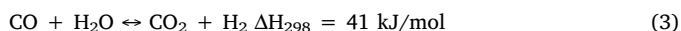
The mass balance of components (*i*: H₂, CO₂, CO, methanol, and H₂O) over a differential control volume is represented by the following sets of ordinary differential equations [31]:

$$W \frac{d\omega_i}{dV} = u_A \tilde{R}_i(x) \quad (1)$$

It is presumed that *W* (total mass flow rate) does not vary with mole number change, reactor temperature, and pressure gradient along the path. In Eq. (1), the mass fractions (ω_i) vary along the path due to reactions. u_A is the catalyst activity and varies between 0 and unity.

The methanol synthesis reactions from a mixture of hydrogen and CO₂ are: the hydrogenation of CO₂, Eq. (2), and the reverse water gas shift (RWGS) reaction, Eq. (3).





Reaction (2) is exothermic while Reaction (3) is endothermic.

The reaction rate of CO_2 hydrogenation to methanol, Reactions (2) and (3), over copper-zinc oxide on alumina (CuZA) catalyst were taken from [32]. The reaction rates are functions of component mole fractions, and reactor pressure and temperature.

$$r_2 = \frac{k_2 K_{\text{CO}_2} (P_{\text{CO}_2} P_{\text{H}_2}^{\frac{3}{2}} - \frac{P_{\text{CH}_3\text{OH}} P_{\text{H}_2\text{O}}}{K_{p_2} P_{\text{H}_2}^{\frac{3}{2}}})}{(1 + K_{\text{CO}} P_{\text{CO}} + K_{\text{CO}_2} P_{\text{CO}_2}) (P_{\text{H}_2}^{\frac{1}{2}} + \frac{K_{\text{H}_2\text{O}} P_{\text{H}_2\text{O}}}{K_{p_3} P_{\text{H}_2}^{\frac{1}{2}}})} \quad (4)$$

$$r_3 = \frac{k_3 K_{\text{CO}_2} (P_{\text{CO}_2} P_{\text{H}_2} - \frac{P_{\text{H}_2\text{O}} P_{\text{CO}}}{K_{p_3}})}{(1 + K_{\text{CO}} P_{\text{CO}} + K_{\text{CO}_2} P_{\text{CO}_2}) (P_{\text{H}_2}^{\frac{1}{2}} + \frac{K_{\text{H}_2\text{O}} P_{\text{H}_2\text{O}}}{K_{p_3} P_{\text{H}_2}^{\frac{1}{2}}})} \quad (5)$$

Where:

$$k_2 = 23.4 \times \exp\left(\frac{-52570}{RT}\right) \quad k_3 = 4.84 \times 10^{11} \times \exp\left(\frac{-140020}{RT}\right)$$

$$K_{p_2} = 1.959 \times 10^{-11} \times \exp\left(\frac{59030}{RT}\right) \quad K_{p_3} = 117.52 \times \exp\left(\frac{-39652}{RT}\right)$$

$$K_{\text{CO}_2} = 7.05 \times 10^{-7} \times \exp\left(\frac{61700}{RT}\right) \quad K_{\text{CO}} = 2.16 \times 10^{-5} \times \exp\left(\frac{46800}{RT}\right)$$

$$\frac{K_{\text{H}_2\text{O}}}{K_{p_3}^{\frac{1}{2}}} = 6.37 \times 10^{-9} \times \exp\left(\frac{84000}{RT}\right)$$

For each chemical component (H_2 , CO_2 , CO , methanol, and H_2O) a net production rate (positive if it is produced and negative if it is consumed considering all the reactions in the system) is defined:

$$R_{\text{H}_2} = -3r_1 + r_2 \quad (6)$$

$$R_{\text{CO}_2} = -r_1 + r_2 \quad (7)$$

$$R_{\text{CO}} = -r_2 \quad (8)$$

$$R_{\text{MeOH}} = r_1 \quad (9)$$

$$R_{\text{H}_2\text{O}} = r_1 - r_2 \quad (10)$$

The fluid model is on mass basis ($\text{kg}/\text{m}^3/\text{s}$) so it is required that Eqs. (6)–(10) be multiplied by molecular weights (M_i), i.e. $\tilde{R}_i = R_i M_i$. At any point along the reactor, the sum of reaction rates on mass basis ($\tilde{R}_i$) must be zero, i.e. $\sum_{i=1}^5 \tilde{R}_i = 0$.

The energy conservation law of the fluid is written as follows [31]:

$$W c_p \frac{dT}{dV} = \sum_{j=1}^2 (-\Delta_r H_j) r_j - U a (T - T_w) \quad (11)$$

$$c_p = \sum_{i=1}^5 c_{p,i} \omega_i, \quad a = \frac{\text{Area}}{V} \quad (12)$$

Where, c_p is the heat capacity of the gas mixture ($\text{kJ}/\text{kg K}$), $(-\Delta_r H_j)$ is the reactions heat (kJ/kmol), U is the overall heat transfer coefficient, a is the area density ($\text{kW}/\text{m}^2 \text{K}$), T is the reactor temperature (K), and T_w is the coolant medium temperature (K).

The optimal design functions and concentration of the key component i.e. methanol at the end of reactor path is found by solving the set of mass and energy ordinary differential equations Eq. (13). The objective function (J) is a function of state variables $z = [\omega_i^T, T]^T$ at the end of reactor path.

$$\max J_s. \text{ t. } \frac{dz}{dV} = f(z, \text{DOFs}); \quad z(0) = z_0 \quad (13)$$

Where

$$z = [\omega_i^T, T]^T$$

The optimization problems consisting of ordinary or partial differential equations are called optimal control problems. In this paper, the collocation method on finite elements (direct method) was used to transform the ordinary differential equations (ODEs) to nonlinear algebraic equations.

Partial differential equations represented by Eqs. (1) and (13) were discretized on volume (V) and the resulting nonlinear equations were included in the optimization problem.

The following objective functions were maximized to find the optimal decision variables:

$$J_1 = \text{Methanol mass production rate at the end of path (kg/s)} \quad (14)$$

$$J_2 = \text{CO}_2 \text{ conversion (\%)} \quad (15)$$

$$J_3 = \text{H}_2 \text{ conversion (\%)} \quad (16)$$

Objective function (14), the mass production rate of methanol at the effluent of the reactor (reactor productivity), is the maximum attainable production rate of the key component at the end of path subject to the optimization constraints. The MATLAB optimization toolbox finds the optimal decision variables of the non-linear optimization problem.

Objective function (21) is the maximum possible CO_2 utilization (or the minimum CO_2 release to the air) along the reactor path. CO_2 is converted to CO and methanol. So, maximizing CO_2 utilization does not necessarily lead to the maximum methanol production.

Objective function (16) describes the maximum hydrogen conversion potential (or the maximum energy storage) of the methanol reactor.

4. Results and discussion

4.1. Methanol production from CO_2 hydrogenation (Fig. 2a)

Table 1 shows the various cases with different syngas composition (H_2 to CO_2 ratio), and objective function criteria (Eqs. (14)–(16)).

4.1.1. Maximizing methanol mass production rate

Case (1). The CO_2 and hydrogen streams are fed to a single reactor with

Table 1

Optimization results of Fig. 2a with different objective function criteria (Eqs. (14), (15), and (16)), and feed composition. The reactor volume for all cases is 100 m^3 .

Case #	Objective function criteria	Molar ratio of H_2/CO_2 at the inlet	W_0 (kg/s)	ω_{MeOH}^1 (%)	Methanol production rate (kg/s)	CO_2 conversion (%)	H_2 conversion (%)	Maximum reactor temperature ($^\circ\text{C}$)	Coolant temperature ($^\circ\text{C}$)	Average Ua ($\text{kW}/\text{m}^3\text{K}$)
Case (1)	(14)	2	34.7	7.02	10.95	48.9	64.0	269	265	90.07
Case (2)		3	36.1	12.12	17.32	72.3	79.5	270	265	109.78
Case (3)	(15)	2	34.7	7.05	10.97	49.0	64.1	270	267	135.10
Case (4)		3	36.1	12.29	17.41	72.6	79.8	270	267	164.68
Case (5)	(16)	2	34.7	7.05	10.97	49.0	64.1	270	267	90.07
Case (6)		3	36.1	12.29	17.41	72.6	79.8	270	267	109.78

1: Weight % of methanol at the outlet of the reactor.

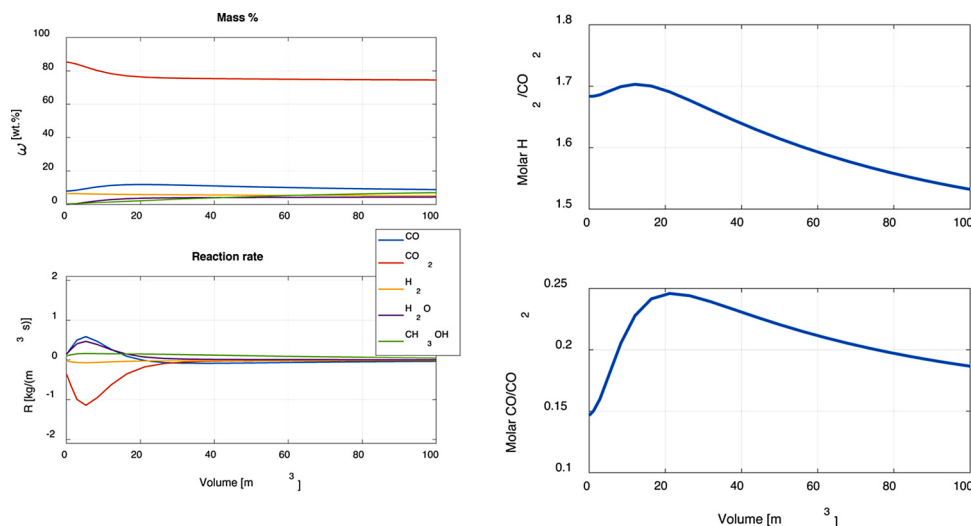


Fig. 3. a: Mass percentages profiles and reaction rates of components of Case (1) b: Change of molar H₂/CO₂ and CO/CO₂ along the single-stage reactor.

volume of 100 m³ [31]. The inlet mass flowrate is 34.7 kg/s. The optimization problem has 3 degrees of freedom. The methanol weight percentage at the reactor outlet is 7.02 %. The methanol production rate of this case is lower than that reported in Ref. [31] where methanol production from a CO-rich syngas stream was considered. This owes to the fact that the inlet mole fraction of CO to the reactor is 21 % in Ref. [31]. The conversion of hydrogen is 64.0 % and about 48.9 % of CO₂ is converted to methanol and CO. The conversions of hydrogen and CO₂ are higher than those reported in Refs. [12,13] because of impurity-free syngas feed, higher reactor pressure and temperature. The catalyst activity is unity and Ua is 90.07 kW/m³K. The weight percentage of CO at the end of reactor is 8.85 %, and the maximum CO weight percentage is 11.92 % and takes place at the 21.1 % of the reactor volume. The change of mass fraction of components and reaction rates along the reactor are shown in Fig. 3a. As evident from this figure, CO is first produced in the 0–21.12 % of the reactor volume and then consumed along the reactor. CO₂ and H₂ reaction rates are negative, and water and methanol reactions rates are always positive along the reactor. Fig. 3b represents the change of molar H₂/CO₂ and CO/CO₂ ratios along the reactor. The molar H₂/CO₂ ratio increases from 1.68 (at the inlet) to about 1.70 and then declines to 1.53 at the end of reactor. The CO/CO₂ increases from 0.15 to about 0.25 at 21 % of the reactor volume and then declines to 0.19. The carbon efficiency of the process is 47.37 % which is lower than that reported in Ref. [31] since the inlet mole fraction of CO to the CO₂ hydrogenation reactor is zero.

Case (2). In this case, the CO₂ and hydrogen streams with a H₂/CO₂ molar ratio of 3 are fed to a single reactor. The inlet mass flowrate to the reactor is 36.1 kg/s. There are 3 degrees of freedom for optimization. The methanol mass production rate at the reactor outlet is 17.32 kg/s with weight percentage of 12.12 %. Increasing the inlet H₂/CO₂ molar ratio from 2, Case (1), to 3 increases the methanol production rate by 58.17 %. The conversion of hydrogen is 79.5 % and about 72.3 % of CO₂ is converted to methanol and CO. The catalyst activity is unity and Ua is 109.78 kW/m³K which is higher than Case (1). The weight percentage of CO at the end of reactor is 11.08 %, the maximum CO weight percentage is 16.00 % and takes place at the 21.12 % of the reactor volume. The profiles are shown in Fig. 4a and b. The H₂/CO₂ increases from 4.6 at the beginning of the reactor to 5.28 and then goes down to 5.14 at the reactor outlet. The reactor CO/CO₂ ratio hits the maximum value of 0.42 and drops to 0.31 at the reactor outlet.

4.1.2. Maximizing CO₂ utilization (conversion)

Case (3). This case is similar to Case (1) except that the objective function is maximization of CO₂ utilization (conversion) at the end of reactor. The methanol mass fraction at the outlet of the reactor is 7.05 % that is higher than that of Case (1). The conversion of hydrogen is 64.1 % and about 49.0 % of CO₂ is converted to methanol and CO. The catalyst activity is unity and Ua is 135.10 kW/m³K which is higher than Case (1). The weight percentage of CO at the end of reactor is 8.9 %, and the maximum CO weight percentage is 12.2 %. The maximum H₂/CO₂ molar ratio is 1.71 and the maximum CO/CO₂ ratio is 0.25.

Case (4). Here, the hydrogen to CO₂ molar ratio is 3 and is fed to a single reactor and the CO₂ conversion was maximized. The inlet mass flowrate to the reactor is 36.1 kg/s. The mass flowrate of methanol at the outlet of the reactor is 17.41 kg/s with weight percentage of 12.29 %. Increasing the inlet H₂/CO₂ molar ratio from 2, Case (3), to 3 increases the methanol production rate by 58.71 %. The conversion of hydrogen is 79.8 % and about 72.6 % of CO₂ is converted to methanol and CO. The CO₂ conversion of this case is 48.16 % which is higher than Case (3). The catalyst activity is 1 and Ua is 164.68 kW/m³K which is lower than Case (3). The weight percentage of CO at the end of reactor is 10.93. The maximum CO/CO₂ ratio is 0.42 and the maximum H₂/CO₂ molar ratio is 5.28.

4.1.3. Maximizing H₂ conversion

Case (5). In this case, the CO₂ and hydrogen with molar ratio of 2 are fed to a single reactor. The inlet mass flowrate to the reactor is 34.7 kg/s. The mass fraction of methanol at the outlet of the reactor, the H₂ and CO₂ conversions are very close to those of Case (1). In addition, the coolant temperature and maximum reactor temperature are close to those of Case (1) where the methanol concentration at the end of reactor was maximized. In this case, the catalyst activity is unity and the Ua is 90.07 kW/m³K. The weight percentage of CO at the end of reactor is 8.90 %. The maximum H₂/CO₂ molar ratio is 1.70 and the maximum CO/CO₂ ratio is 0.25.

Case (6). This case is similar to Case (2) except that the objective function was set to maximize the H₂ conversion. The optimization results of this case are very close to those of Case (2). In addition, the production rate of methanol is greater than Case (5) where the H₂/CO₂ molar ratio is 2. The catalyst activity is unity and Ua is 109.78 kW/m³K. The maximum H₂/CO₂ molar ratio is 5.28 and the maximum CO/CO₂ ratio is 0.42.

A case study was conducted to investigate the effect of feed

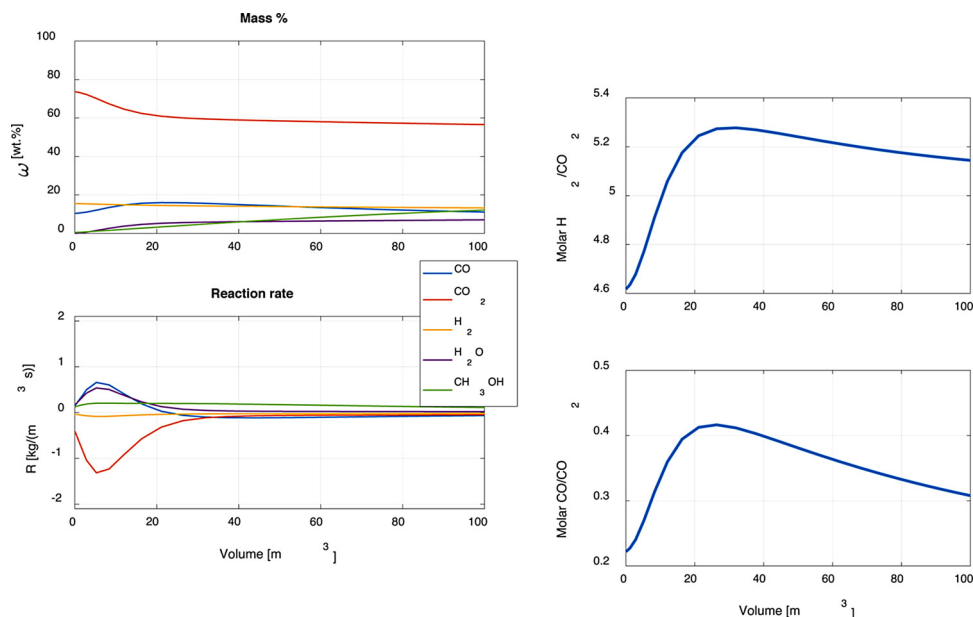


Fig. 4. a: Mass percentages profiles and reaction rates of components of Case (2) b: Change of molar H₂/CO₂ and CO/CO₂ of Case (2).

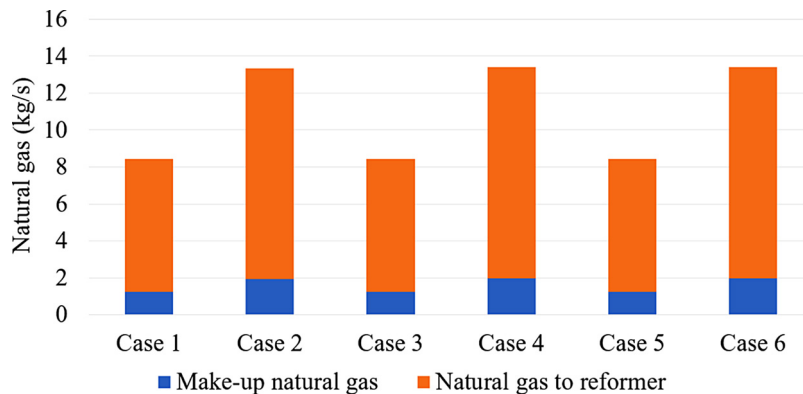


Fig. 5. Natural gas consumption of Case (1)- Case (6) in the conventional methanol production route.

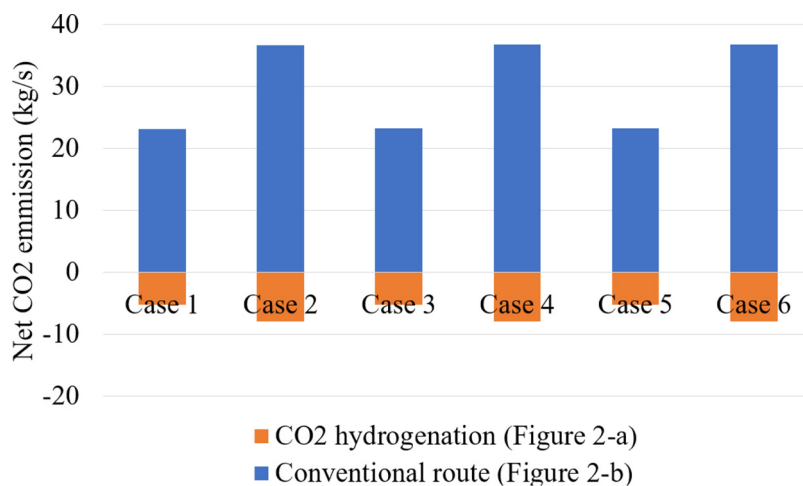


Fig. 6. Comparison of CO₂ emission of CO₂ hydrogenation (Fig. 2a) with conventional methanol production (Fig. 2b) routes.

temperature on the optimal decision variables of the once-through reactor path. A feed containing CO₂ and hydrogen with molar ratio of H₂/CO₂ of 2 (and inlet mass flowrate to the reactor of 34.7 kg/s) was considered. For a single-stage reactor and Eq. (14) as the objective function, increasing the feed temperature from 190 to 250 °C results in

increasing the production rate of methanol by 0.55 %, and decreasing the coolant temperature from 240.1–236.7 °C. In addition, the catalyst activity is equal to unity for all cases and Ua declines from 59.80 to 12.98 kW/m³K.

Another case study was carried out to investigate the impact of

staging of the methanol reactor on the methanol production rate. For a two-stage methanol reactor and the inlet H_2/CO_2 ratio of 2, the reactor productivity increases by 0.57 % compared to Case (1). The volume of the first reactor becomes 10.8 m^3 . In addition, the average Ua of the system reduces to $25.84\text{ kW/m}^3\text{K}$. The conversions of CO_2 and H_2 of the two-stage reactor system are 49 % and 64.2 %, respectively.

For a two-stage methanol reactor and the inlet H_2/CO_2 ratio of 3, the methanol production rate of the staged reactor is system is 0.7 % higher than the single-stage case, i.e. Case (2). The volume of the first reactor becomes 6.9 m^3 and the average Ua of the system is the same as the Ua of Case (2). The conversions of CO_2 and H_2 of the staged reactor system are 61.2 %, and 66.6 %, respectively.

4.2. Conventional methanol production (Fig. 2b)

The natural gas demand in the conventional methanol production route is 28.5 MMBTU/tonne methanol which is equivalent to 0.0343 MMSCFD/tonne methanol [33]. The required natural gas to the process side and to the fired heater (Fig. 2b) were calculated based on the methanol production rates of Case (1)- Case (6) and is shown in Fig. 5. The steam to carbon ratio to the reformer was fixed at 1 in Aspen Hysys. The make-up natural gas to the fired heater supplies the required heat to preheat feed streams and to generate steam to drive compressors in the air separation unit. The required natural gas to the reformer (process side) of Fig. 2b is about 85 % of the total natural gas consumption of the process with methanol production rate of the cases represented in Table 1. For each objective function criteria, as evident from Fig. 5, the higher the H_2/CO to the methanol reactor, the higher the required natural gas is. For instance, the natural gas consumption of Case (2) > Case (1), Case (4) > Case (3), and Case (6) > Case (5). In addition, for each case, the natural gas consumption in the process side is higher than the make-up natural gas.

In the case that the hydrogen intake to the methanol reactor is supplied from electrolyzing water, a pure oxygen stream is available to be sold or to be used in other processes. At H_2/CO_2 molar ratio of 2, the produced oxygen is 722.35 mol/s while about 1083.50 mol/s pure oxygen is available at H_2/CO_2 molar ratio of 3. The required energy to produce 1 t oxygen via conventional air separation unit is 400 kW h power [24]. This means that producing 1 t of pure oxygen (Fig. 2b) requires ca. 0.08294 t natural gas.

In terms of CO_2 emissions, the process shown in Fig. 2a uses a pure CO_2 stream from a CO_2 capturing unit. In other words, no fossil fuel is used in this configuration and the pure oxygen stream from the electrolyzer saves 1.92 kg/s natural gas when the inlet H_2/CO_2 is 2 and ca. 2.88 kg/s natural gas in the cases with the inlet H_2/CO_2 of 3. The higher the natural gas savings, the lower the CO_2 emissions. Thereby, the net CO_2 emissions of configuration Fig. 2a is -5.27 kg/s in Case (1), Case (3), and Case (5) while it is -7.91 kg/s in Case (2), Case (4), and Case (6). However, Fig. 2b utilizes natural gas in the process side and in the fired heater which in turn emits CO_2 to the environment. Fig. 6 compares the net CO_2 emissions of conventional methanol production route when the same amount of methanol of Case (1)- Case (6) is produced. As evident from this figure, in the case that hydrogen is produced via water electrolysis, the CO_2 emission of the process route shown in Fig. 2a is negative. Higher inlet molar H_2/CO_2 ratio to the methanol reactor translates to higher oxygen production via electrolyzer and higher natural gas saving compared to the conventional air separation route. In the conventional methanol production route, the higher the inlet H_2/CO_2 ratio, the higher the CO_2 emission.

5. Conclusions

The methanol production via conventional and CO_2 hydrogenation routes are investigated. In the CO_2 hydrogenation route, methanol production rate, CO_2 conversion, and hydrogen conversion are maximized. Based on the optimization results, the methanol production rate

of the cases with inlet H_2/CO_2 of 3 is higher than those with inlet H_2/CO_2 composition of 2.

The impact of feed temperature on the optimal decision variables of the reactor path was investigated for the case that the H_2/CO_2 ratio of feed is 2. Altering the feed temperature between 190 and $250\text{ }^\circ\text{C}$ increases the methanol mass fraction by 0.55 % and reduces the coolant temperature from $240.1\text{--}236.7\text{ }^\circ\text{C}$. In addition, the heat transfer area drops from 59.80 to $12.98\text{ kW/m}^3\text{K}$.

Key performance indicators methanol production via conventional and CO_2 hydrogenation routes are compared. For each objective function criteria, the higher the H_2/CO to the methanol reactor, the higher the required natural gas is. For instance, the natural gas consumption of Case (2) > Case (1), Case (4) > Case (3), and Case (6) > Case (5). If hydrogen is produced via water electrolysis, the CO_2 emission of the process route shown in Fig. 2a is negative. In the conventional methanol production route, for each objective function criteria, the higher the inlet H_2/CO_2 ratio the higher the CO_2 emission.

6. CRediT author statement

All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be submitted to or published in any other publication before its appearance in the *Journal of Environmental Chemical Engineering*.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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